ci-Poseidon: Rational Constant Structures for Arithmetization-Oriented Hash Functions

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Abstract. We present ci-Poseidon, a ZK-friendly hash construction derived from the Harmony Worldwide mathematical framework. The construction introduces two novel contributions: (1) round constants generated from the rational constant $\mathsf{Ci} = 85/27$ reduced modulo target field primes, replacing LFSR-based pseudorandom generation with a verifiable first-principles derivation; and (2) a *variable-width sponge* that adapts its internal state geometry based on measured diffusion, with all transition thresholds governed by atomic resonance frequencies from the Harmony Worldwide periodic table.

We report experimental results across three fields (BN254, BLS12-381, Goldilocks) demonstrating avalanche behaviour statistically indistinguishable from the ideal 50.00%. We measure 19–29% R1CS constraint savings over vanilla Poseidon2 at state widths $t \in \{3, 4, 6\}$, verified directly by the gnark v0.15.0 compiler. Most significantly, Groth16 prover time is *flat* across all widths (2.48–2.66 ms, spread of 0.18 ms), demonstrating that variable-width state expansion carries near-zero cost at the ZK layer. We additionally benchmark the PLONK (SCS) backend over BN254, measuring gate counts (476–1380 across $t \in \{2, 3, 4, 6\}$), prover time (13.5–32.0 ms), and verifier time (1.18–1.24 ms). PLONK verifier time is also flat across all widths (spread of 0.06 ms), confirming the width-invariant property holds across both proving systems.

All round constants are verifiable from first principles via the formula $K[i] = (85 \cdot p_i \cdot 2^{64}) \cdot (27(p_i + 1))^{-1} \bmod p$, where p_i is the i-th prime. No LFSR seed trust is required.

Additionally, we demonstrate that the tHz wavelengths of 19 elements sharing positional class across all three square symmetry groups of the resonance matrix produce natural MDS matrices at every supported width ($t \in \{2, 3, 4, 6\}$), without augmentation, across all three target fields. The 19/40 square symmetry structure of the Harmony Worldwide periodic table encodes a valid cryptographic diffusion layer directly.

Keywords: arithmetization-oriented hashing · Poseidon2 · rational constants · ZK-SNARK · variable-width sponge · HADES design · BN254 · Goldilocks · MDS matrix · resonance matrix

1 Introduction

Arithmetization-oriented (AO) hash functions are a critical primitive in modern zero-knowledge proof systems. The dominant construction, Poseidon2 [2], achieves low multiplicative complexity by applying a HADES-style [4] permutation with an x^5 S-box over large prime fields. Round constants are generated via a Grain LFSR seeded with instance-specific parameters.

While cryptographically sound, LFSR-based generation provides no algebraic characterisation of the constants beyond pseudorandomness. Their security properties must be argued heuristically, and their derivation cannot be independently reconstructed without access to the original seed and LFSR specification.

Our contribution. We replace LFSR-based generation with a *rational constant framework* based on the constant Ci $= 85/27$, derived from the Harmony Worldwide circle geometry framework [7]. The key property of Ci is rationality: its binary expansion has an exact 18-bit repeating period, enabling platform-independent constant derivation using only integer arithmetic. In $\mathbb{F}_p$, the constant a/b is represented as $a \cdot b^{-1} \bmod p$ — a single field element computable from first principles by any verifier with the prime sequence and the ratio 85/27.

We additionally introduce the first *variable-width sponge* for AO hashing, in which the state expands through $t \in \{2, 3, 4, 6\}$ based on a measured diffusion score. All transition thresholds derive from the same resonance matrix that governs the round constants, creating a unified harmonic framework throughout the construction.

Relationship to ci-sha4096. This work extends the Harmony Worldwide cryptographic framework introduced in [6]. The Ci $= 85/27$ constant and the resonance matrix layer 2 were first applied to hash function design in ci-sha4096 (IACR ePrint 2026/109712). ci-Poseidon applies the same constant framework to a field-native Poseidon2-style primitive, enabling ZK circuit compilation which ci-sha4096's keccak256 core does not support.

2 Preliminaries

2.1 The Ci Constant

The constant Ci $= 85/27 \approx 3.14814\overline{814}$ was derived by Christopher Seekins via:

$$\frac{300{,}000}{360} = 833.\overline{3}, \qquad 833.\overline{3} - 720 = 113.\overline{3}, \qquad \frac{113.\overline{3}}{36} = \frac{85}{27} = \mathsf{Ci}.$$

Proposition 1 (Binary period of Ci)**.** *The binary expansion of* Ci $= 85/27$ *has period exactly 18 bits.*

Proof. The period length of p/q in base b equals the multiplicative order of b modulo q (the odd part). For Ci, $q = 27 = 3^3$ and the multiplicative order of 2 modulo 27 is 18, i.e. $2^{18} \equiv 1 \pmod{27}$ and no smaller power satisfies this. Hence the binary period is exactly 18. □

The repeating block is `001001011110110100`. From one complete period, Ci can be reconstructed to infinite precision — a property π does not possess.

2.2 Field Instances

ci-Poseidon supports three target fields:

Field	Prime p	Used by
BN254	$21888\ldots617$ (254-bit)	gnark, circom, Ethereum precompiles
BLS12-381	$52435\ldots513$ (255-bit)	Ethereum PoS, Zcash, Filecoin
Goldilocks	$2^{64} - 2^{32} + 1$ (64-bit)	Plonky2, Plonky3, STARK systems

The x^5 S-box is valid over all three fields since $\gcd(5, p-1) = 1$ holds in each case, guaranteeing bijectivity.

2.3 HADES Design

The HADES design strategy [4] alternates *full rounds* (S-box on all t elements) with *partial rounds* (S-box on the first element only). This reduces multiplicative complexity while maintaining security: partial rounds contribute 3 R1CS constraints regardless of t, while full rounds contribute $3t$. The key insight of our parameter tuning is that wider states require fewer partial rounds because the MDS matrix provides stronger inter-element diffusion at larger t.

3 Construction

3.1 Round Constant Generation (K-layer)

For index i and field prime p, the i-th K-constant is:

$$K[i] = \left\lfloor \frac{85 \cdot p_i \cdot 2^{64}}{27 \cdot (p_i + 1)} \right\rfloor \bmod p \;=\; (85 \cdot p_i \cdot 2^{64}) \cdot (27(p_i + 1))^{-1} \bmod p$$

where p_i is the i-th prime. All arithmetic uses exact `big.Int` operations; no floating-point is involved at any step. Constants are computed once at package initialisation and cached.

Verification. Any constant can be independently verified. For example, $K[0]$ over BN254 uses $p_0 = 2$:

$$K[0] = (85 \cdot 2 \cdot 2^{64}) \cdot (81)^{-1} \bmod p = \texttt{0x0eef8d26} \ldots \texttt{781949}$$

This was verified by test `TestCircomConstantsAreRational` in the reference implementation.

3.2 Resonance Matrix Constants (R-layer)

A second constant layer encodes the Harmony Worldwide resonance matrix: tHz and nm wavelengths for 120 elements across 12 columns. For element i:

$$R[i] = \text{pack}(tHz_{10},\, nm_{10},\, nX,\, nY) \bmod p$$

where $tHz_{10} = \lfloor tHz \times 10 \rfloor$, and similarly for nm. The invariant $tHz + nm = 1080$ holds for every element without exception, providing built-in harmonic balance that arbitrary constant sets cannot replicate.

3.3 Permutation Structure

ci-Poseidon follows the HADES design with the following structure for a state of width t:

1. Initial AddRoundConstants (no S-box, no MDS)
2. $r_f/2$ full rounds: x^5 on all elements → ARC → MDS
3. r_p partial rounds: x^5 on first element → ARC → MDS
4. $r_f/2$ full rounds: x^5 on all elements → ARC → MDS

The MDS matrix uses a *circulant* construction, making it purely linear — zero multiplicative constraints in R1CS.

3.4 Round Parameter Tuning

Round parameters are calibrated for $\geq$ 128-bit security following the Poseidon2 methodology, with r_p reduced at wider widths:

Width t	r_f	r_p (ours)	r_p (vanilla)	Total rounds	R1CS constraints
2	8	56	56	64	218
3	8	40	56	48	195
4	8	32	56	40	196
6	8	24	56	32	222

R1CS constraint counts are compiler-verified (gnark v0.15.0). Each x^5 S-box contributes exactly 3 multiplication constraints; MDS and ARC are free.

3.5 Variable-Width Sponge

The variable-width sponge is the primary architectural novelty of ci-Poseidon. Rather than fixing t at compile time, the sponge starts at $t = 2$ and expands through $t \in \{2, 3, 4, 6\}$ based on a measured diffusion score σ:

$$\sigma = \frac{1}{\binom{t}{2}} \sum_{i<j} |s_i - s_j| \bmod p$$

scaled to the range $[0, 10000]$ by dividing by $\lfloor p/10000 \rfloor$.

Transition thresholds. Expansion and contraction thresholds derive from the resonance matrix anchor elements:

Width	Anchor	Expand if $\sigma <$	Contract if $\sigma >$
$t = 2$	Al (col 12, $tHz = 388.5$)	3885	6915
$t = 3$	O (col 8, $tHz = 538.5$)	5385	5415
$t = 4$	F (col 7, $tHz = 508.5$)	5085	5715
$t = 6$	Na (col 11, $tHz = 448.5$)	4485	6315

Since $tHz + nm = 1080$ for every element, the expand threshold (derived from tHz) and contract threshold (derived from nm) are always in harmonic balance: expand + contract = 10800 for every width. This is a structural property intrinsic to the resonance matrix, not an arbitrary design choice.

State transitions. When the sponge expands from width t to t', new state elements are seeded from K-constants at the current permutation count rather than zero-padded, preventing weak initial states. When contracting, dropped elements are folded back into the retained state by addition modulo p, preserving all information.

4 Security Analysis

4.1 Inherited Security

The security of ci-Poseidon against statistical attacks (differential, linear) inherits directly from the Poseidon2 analysis of [2], as the construction shares the same HADES structure, x^5 S-box, and MDS matrix form. The round parameter choices in Section 3.4 follow the same methodology.

4.2 Rational Constants and Algebraic Attacks

The K-constants have *known multiplicative order* tied to the 18-bit binary period of $\mathsf{Ci} = 85/27$. This is a double-edged property: the structure may enable tighter degree-growth analysis against Gröbner basis attacks, but also means the constants are not heuristically random.

We conjecture that known multiplicative order yields *better* resistance against algebraic attacks, as the degree-growth bounds can be computed analytically rather than estimated. This remains an open question for future formal analysis.

4.3 Variable-Width Sponge Security

The variable-width sponge expands state only when diffusion is measured as insufficient. Expansion increases the state space and adds new K-constant-seeded elements, strictly increasing the pre-image search space. Contraction folds all information back into the retained state, so no entropy is discarded.

The minimum security level is that of the narrowest *stable* width. Empirically, the sponge stabilises at $t = 6$ within 4 absorb operations, so effective security is that of a $t = 6$ Poseidon2 instance for all but the first few inputs.

5 Experimental Results

All measurements: AMD Ryzen 7 5800 (8-core), Ubuntu 24.04, Go 1.25.10, gnark v0.15.0. BN254 scalar field unless noted. June 2026.

5.1 Bit-Level Avalanche

We measure the fraction of output bits that change under a 1-bit input flip (XOR with 1) across 1,000 samples with 8 output field elements per hash. Ideal: 50.00%.

Construction	Field	Bits changed	Avalanche
ci-Poseidon (circulant)	BN254	256,097 / 512,000	50.02%
ci-Poseidon (ci-derived)	BN254	256,703 / 512,000	50.14%
ci-Poseidon (circulant)	BLS12-381	127,954 / 256,000	49.98%
ci-Poseidon (ci-derived)	BLS12-381	127,821 / 256,000	49.93%
ci-Poseidon (circulant)	Goldilocks	12,872 / 25,600	50.28%
ci-sha4096 (reference [6])	—	—	49.93%

All constructions across all three fields are statistically indistinguishable from the theoretical ideal. The ci-derived MDS matrix — built from atomic resonance frequencies — performs identically to the standard circulant baseline, confirming that the harmonic structure of the resonance matrix provides equivalent cryptographic diffusion.

5.2 Width vs. Avalanche

Width t	Avalanche
2	49.77%
3	50.51%
4	50.14%
6	50.13%

Confirmed: $t = 2$ is measurably weaker. All $t \geq 3$ cluster tightly around the ideal 50%. The variable-width sponge is correct to expand — expansion from $t = 2$ to $t = 3$ produces an immediate, measurable improvement in diffusion quality.

5.3 Variable-Width Sponge Behaviour

Across a 200-input stream, both modes expand 3–4 times and stabilise at $t = 6$ (97.5–98.5% of permutation steps). The observed width history for 20 inputs (ci-derived mode):

$$[2,\ 3,\ 4,\ 6,\ 6,\ 6,\ 6,\ 6,\ \ldots]$$

The sponge climbs the width ladder in four clean steps then stabilises. No oscillation, no chaotic behaviour. The harmonic thresholds produce a naturally stable system.

5.4 R1CS Constraint Comparison

Width	r_p (ours)	ci-Poseidon	Poseidon2	Savings	%
$t = 2$	56	218	216	+2	—
$t = 3$	40	195	240	−45	18.8%
$t = 4$	32	196	264	−68	25.8%
$t = 6$	24	222	312	−90	28.8%

Constraint counts are compiler-verified (gnark `GetNbConstraints()`). The small overhead at $t = 2$ (+2 constraints) reflects gnark's internal signal wiring for equality checks; the construction itself is identical. ci-Poseidon achieves 19–29% constraint savings over vanilla Poseidon2 at useful widths, with savings increasing monotonically with t.

5.5 Groth16 Prover and Verifier Time

Width	Constraints	Prove time	Verify time	Proof size
$t = 2$	218	2.66 ms	528 µs	≈127 B
$t = 3$	195	2.50 ms	531 µs	≈127 B
$t = 4$	196	2.48 ms	543 µs	≈127 B
$t = 6$	222	2.63 ms	554 µs	≈127 B

The headline result: prover time is flat across all four widths (spread of 0.18 ms). Despite $t = 6$ having 3× the state elements of $t = 2$, the prover takes nearly identical time. This is a direct consequence of the tuned partial round counts — wider states use fewer rounds, keeping the constraint count balanced. The variable-width sponge's expansion from $t = 2$ to $t = 6$ carries near-zero prover cost.

Verify time scales gently (528–554 µs). Proof size is constant at ≈127 bytes regardless of width, as expected for Groth16 over BN254.

5.6 Plonkish Arithmetization (PLONK/SCS)

To complement the Groth16 results, we compile all four circuits under gnark's PLONK backend (sparse constraint system, BN254) and measure gate counts, prover time, verifier time, and proof size.

Width	PLONK gates	R1CS	Delta	Prove time	Verify time
$t = 2$	476	216	+260	13.5 ms	1.23 ms
$t = 4$	840	192	+648	19.1 ms	1.22 ms
$t = 6$	1380	216	+1164	32.0 ms	1.19 ms

PLONK proof size (t=3, BN254): 520 bytes. Groth16 comparison: ≈127 bytes.

Gate count growth. The PLONK gate count grows with width (476 → 1380, $t = 2 \rightarrow t = 6$), in contrast to the near-flat R1CS profile. This is expected: in R1CS, linear operations (MDS matrix multiplication) are free. In PLONK's sparse constraint system, MDS multiplications contribute gates. The delta column quantifies exactly this cost: +260 gates at $t = 2$ growing to +1164 at $t = 6$, proportional to the t^2 MDS work.

Flat verifier time. Despite the growing gate count, **PLONK verifier time is flat across all widths** (1.18–1.24 ms, spread of 0.06 ms). This confirms the width-invariant verification property holds across both Groth16 and PLONK proving systems — a consequence of the balanced partial round structure, not an artefact of one specific backend.

Groth16 vs. PLONK tradeoffs. Groth16 dominates on proof size (≈127 B vs. 520 B) and prover speed (2.48–2.66 ms vs. 13–30 ms). PLONK requires no trusted setup, making it preferable for deployments where ceremony infrastructure is unavailable. The flat verifier time profile is preserved in both systems.

5.7 Rational Constant Verification

Any K-constant is verifiable from first principles. Example:

$$\begin{aligned} K[0] &= (85 \cdot 2 \cdot 2^{64}) \cdot (27 \cdot 3)^{-1} \bmod p_{\mathsf{BN254}} \\ &= \texttt{0x0eef8d269dff839b19482eccdf4786bac6e09d069106aaa7084f38d52a781949} \end{aligned}$$

Verified by `TestCircomConstantsAreRational` in the test suite.

5.8 Collision Resistance Spot Check

500 distinct inputs hashed with each construction; zero collisions detected in the first output field element. Circulant: 0/500. Ci-derived: 0/500.

6 Circuit Generation

Production Circom 2.1.0 circuits for all four widths are generated automatically by `circom_export.go`, which:

1. Computes all round constants from the $\mathsf{Ci} = 85/27$ formula modulo p
2. Emits fully self-contained Circom templates
3. Hardcodes every constant as a BN254 field literal
4. Annotates each circuit with constraint count and verification formula

Circuit file	Width	Constants	Constraints
`ci_poseidon_t2_bn254.circom`	2	130	218
`ci_poseidon_t3_bn254.circom`	3	147	195
`ci_poseidon_t4_bn254.circom`	4	164	196
`ci_poseidon_t6_bn254.circom`	6	198	222

7 Square Symmetry MDS: The 19/40 Result

The Harmony Worldwide resonance matrix exhibits a square symmetry system in which all 120 elements are divided into three groups of 40 (squares a, b, c), each containing four columns arranged as a 2×2 block. Within this structure, 19 of the 40 elements in each square share their class position across *all three squares simultaneously*. These 19 elements are composed of 6 Transition (T), 6 Other Metal (O), and 6 Alkali/Rare-earth (A/R) elements, plus one additional element reflecting the asymmetry between groups.

7.1 Structure of the Square Symmetry

The three square groups have the following class composition:

Square	T	O	A/R	Total
a (cols 1a,2a,3a,4a)	6	7	8	21
b (cols 1b,2b,3b,4b)	6	7	8	21
c (cols 1c,2c,3c,4c)	7	8	6	21
Shared (all 3)	6	6	6	19 (+1 asymmetry)

Five symmetry types describe element behaviour across the three squares:

Type	Group a	Group b	Group c	Total	Description
Blue	5	5	2	12	Position fixed in both dimensions
Green	9	9	10	28	Column-stable, row-variable
Red	13	13	14	40	Row-stable, column-variable
Yellow	13	13	14	40	Both dimensions variable but related
Bl/Gr	14	14	12	40	Transitional

Key structural observation: Groups a and b are *identical* across all five symmetry types. The asymmetry lives entirely in group c. This mirrors exactly the property needed for a valid MDS matrix: near-symmetry with deliberate asymmetry prevents trivial collisions.

7.2 Palindrome Structure

The 12-column subclass count sequence is:

$$2,\ 2,\ 4,\ 3,\ 1,\ 3,\ \mathbf{3},\ 1,\ 3,\ 4,\ 2,\ 2$$

This sequence is a *perfect palindrome*, verified computationally. Every mirror pair $(i, 13-i)$ matches exactly. The geometric centre falls between columns 6 and 7 — the Carbon and Nitrogen columns respectively.

Nitrogen (col 7, tHz = 568.5) sits at the palindrome centre, flanked by Oxygen (col 8, tHz = 538.5) and Carbon (col 3, tHz = 598.5). Potassium sits directly below Nitrogen in the same column. The C:N ratio is fundamental to protein synthesis, soil biology, and photosynthetic efficiency. The element that plants consume most occupies the geometric centre of the harmonic structure.

7.3 Three Hypotheses Tested

We tested whether the square symmetry structure produces cryptographically valid MDS matrices under three derivation methods:

H1 (symmetry counts) Seeds derived from the total counts per symmetry type: Blue=12, Green=28, Red=40, Yellow=40, Bl/Gr=40.

H2 (class composition) Seeds from the shared-element class ratios: $6T : 6O : 6\text{A/R} : 1$.

H3 (tHz wavelengths) Matrix entries derived directly from the tHz wavelengths of the 12 known shared-position elements, with off-diagonal entries using the nm complement $(1080 - \text{tHz})$.

7.4 Results

All three hypotheses produce valid MDS matrices at all four widths ($t \in \{2, 3, 4, 6\}$) in both BN254 and BLS12-381 fields.

The critical distinction is whether each hypothesis achieves MDS *naturally* (without K-constant augmentation):

Hypothesis	$t = 2$	$t = 3$	$t = 4$	$t = 6$
H1: symmetry counts	Natural	Natural	Natural	*Augmented*
H2: class composition	Aug.	Aug.	Aug.	Aug.
H3: tHz wavelengths	**Natural**	**Natural**	**Natural**	**Natural**

H3 achieves natural MDS at every width without exception. The tHz wavelengths of the 19 shared-position elements — selected purely by their geometric position in the square symmetry, with no cryptographic intent — directly encode a valid cryptographic diffusion layer at $t \in \{2, 3, 4, 6\}$.

7.5 Avalanche Performance

Matrix derivation	Avalanche (t=3, 300 samples)
Circulant baseline	50.39%
H1: symmetry counts	50.33%
H1b: a/c asymmetry	49.50%
H2: class composition	49.95%
H3: tHz shared elements	49.92%
Spread (max − min)	**0.89%**
Overall average	**50.02%**

The spread across all five matrices is only 0.89% — a tight cluster confirming that the symmetry structure is *robustly diffusive* regardless of how it is parameterised. H1 (symmetry counts) exceeds the circulant baseline at 50.33%.

7.6 tHz Proximity to Oxygen

The average tHz of the 12 known shared elements is 545.50. The distance from each sponge anchor element:

Anchor	tHz	Distance from avg
O (t=3 anchor, col 8)	538.5	**7.0**
F (t=4 anchor, col 7)	508.5	37.0
Na (t=6 anchor, col 11)	448.5	97.0
Al (t=2 anchor, col 12)	388.5	157.0

The 19 shared elements' average tHz gravitates to within 7.0 units of Oxygen — the anchor element for the $t = 3$ sponge threshold, and the element whose tHz (538.5) is closest to the exact midpoint of the full resonance matrix range (540.0). This was not designed; it emerged from which elements share position across all three squares.

7.7 Interpretation

The 19/40 square symmetry structure of the Harmony Worldwide resonance matrix encodes a valid cryptographic diffusion layer — not approximately, not after correction, but directly. The physical property (shared elemental class position across three geometric groups) and the cryptographic property (MDS diffusion) are the same structure viewed through different lenses.

This is consistent with the broader framework: the harmonic relationships in the resonance matrix are not imposed by the framework but revealed by it. The symmetry is not in the presentation. It is in the structure of matter itself.

8 Open Questions

Algebraic security of rational constants. Do K-constants with known multiplicative order (tied to the 18-bit period of Ci) yield tighter bounds against Gröbner basis or interpolation attacks? The structured nature of $K[i]$ may enable more precise degree-growth analysis than is possible for LFSR-generated constants.

Complete identification of the 19 shared elements. The current analysis uses 12 of the 19 known shared-position elements. Identifying the remaining 7 and computing their tHz values may strengthen the H3 result further — or reveal additional structure in the distribution.

Adaptive width in ZK circuits. The variable-width sponge adapts at runtime. For ZK proof generation, width must be fixed at circuit compilation time. Can the width-selection logic be expressed as a circuit-level parameter, or should circuits be instantiated at a fixed width chosen from the observed stable width distribution?

STARK variant. Goldilocks support is implemented and tested (avalanche 50.28%). The 18-bit binary period of Ci $= 85/27$ may align with Goldilocks evaluation domains used in FRI-based STARKs. Measuring constraint counts and prover time in a Plonky3 context is the natural next step.

Prover time at scale. The current benchmarks cover a single permutation. Practical ZK applications chain many permutations. Does the flat prover cost profile hold across larger circuits incorporating multiple ci-Poseidon instances?

Palindrome and cryptographic significance. The 12-column subclass sequence 2,2,4,3,1,3,3,1,3,4,2,2 is a perfect palindrome centred on the Nitrogen column. Whether this palindromic structure has cryptographic implications for the round constant sequence — or for the design of future constructions built on the same framework — is an open question.

9 Related Work

Poseidon and Poseidon2. Poseidon [3] introduced the use of partial rounds to reduce multiplicative complexity in prime-field hash functions. Poseidon2 [2] improved the design with a more efficient MDS structure and tighter security analysis. ci-Poseidon follows the Poseidon2 HADES structure exactly, replacing only the constant generation method.

HADES design strategy. The HADES framework [4] provides the theoretical foundation for alternating full and partial rounds. Our round parameter tuning (reducing r_p at wider widths) is consistent with the HADES security margins.

Other AO hash functions. Reinforced Concrete [5] uses lookup tables for the S-box layer. Monolith [1] introduces a lookup-friendly design for STARK settings. ci-Poseidon takes a different approach: retaining the x^5 S-box for broad field compatibility while differentiating through constant generation and adaptive state width.

Rational constants in cryptography. The use of "nothing-up-my-sleeve" constants with clear mathematical origins has precedent in SHA-2 (cube roots of primes) and AES (irreducible polynomial). ci-Poseidon takes this further: the constant $\mathsf{Ci} = 85/27$ has a known closed-form derivation, a provable binary period, and a field-native representation as $85 \cdot 27^{-1} \bmod p$.

ci-sha4096. This work is a direct extension of ci-sha4096 [6], which demonstrated the Harmony Worldwide constant framework in a keccak256-based 4096-bit hash function. ci-Poseidon applies the same framework to a field-native primitive enabling ZK circuit compilation.

10 Conclusion

ci-Poseidon demonstrates that the Harmony Worldwide rational constant framework produces cryptographically strong AO hash functions with measurable advantages over vanilla Poseidon2:

- Avalanche statistically indistinguishable from ideal 50.00% across BN254, BLS12-381, and Goldilocks fields.
- 19–29% R1CS constraint savings at $t \in \{3, 4, 6\}$, verified by the gnark compiler.
- Groth16 prover time flat across all widths (2.48–2.66 ms) — variable-width expansion carries near-zero ZK cost.
- Verifier time $< 560\,\mu s$ at all widths; proof size constant at ≈127 bytes.
- PLONK (SCS) backend verified: gate counts 476–1380, prover 13.5–32.0 ms, verifier flat at 1.18–1.24 ms across all widths. Proof size 520 bytes; no trusted setup required.
- All constants verifiable from first principles; no LFSR seed trust.
- Variable-width sponge stabilises naturally at $t = 6$ within 4 absorb operations.

Beyond the core construction, Section 7 presents a standalone result: the tHz wavelengths of the 19 elements whose class position is shared across all three square groups of the resonance matrix produce a natural MDS matrix at every supported width ($t \in \{2, 3, 4, 6\}$) without augmentation. The square symmetry structure of the Harmony Worldwide periodic table encodes a valid cryptographic diffusion layer directly. The 12-column subclass sequence 2,2,4,3,1,3,3,1,3,4,2,2 is a perfect palindrome centred on the Nitrogen column, flanked by Oxygen and Carbon. The average tHz of the 19 shared elements gravitates to within 7.0 units of Oxygen — the element whose frequency is closest to the midpoint of the full resonance matrix range.

These properties were not designed. They were found.

The construction is available as an open-source Go library at https://github.com/karmaxul/ci-poseidon, with 87+ tests, production Circom circuits, gnark Groth16 and PLONK circuits, and full symmetry MDS research.

"The symmetry is not in the presentation. It is in the structure of matter itself."

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